

Total No. of Questions : 8]

SEAT No. :

PB2530

[6263]-428

[Total No. of Pages :2

B.E. (Computer Engineering) (Honors in Data Science)

MACHINE LEARNING AND DATA SCIENCE

(2019 Pattern) (Semester-VII) (410501)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) Explain various types of Hierarchical Clustering [6]

b) Cluster the following dataset using Agglomerative Hierarchical clustering technique- [6]

	X1	X2
A	185	72
B	170	56
C	168	60
D	179	68
E	182	72
F	188	77

Also show intermediate steps.

c) What do you mean by unsupervised learning? Explain in detail. [6]

OR

Q2) a) How to measure the quality of clustering? Explain any three measures. [6]

b) What are different types of partitional clustering? Explain any two of them. [6]

c) What do you mean by Density-Based Spatial Clustering? Explain in detail. [6]

Q3) a) Explain a biological neuron along with its parts. [4]

b) What are the types of artificial neural network? [6]

c) Explain Generalized Delta Learning Rule. [7]

OR

P.T.O.

- Q4)** a) What are the limitations of MLP? [4]
 b) Explain the process of training a perceptron. [6]
 c) How does the learning rate affect the training of the Neural Network?
 What do you mean by Hyperparameters? [7]

- Q5)** a) Illustrate Gradient descent optimization using an example. [6]
 b) Explain Recursive Neural Network. [6]
 c) What are the different types of filters used in CNNs? [6]

OR

- Q6)** a) Does the size of the feature map always reduce upon applying the filters?
 Explain why or why not. [6]
 b) Explain CNN architecture along with diagram. [6]
 c) Explain Recursive Neural Network. [6]

- Q7)** a) Write short note on document representation. [6]
 b) Illustrate tokenization with an example. [6]
 c) What are the practical uses of feature extraction? [5]

OR

- Q8)** a) Write short note on - [6]
 i) Stemming
 ii) Lemmatization
 b) What are various text similarity measures? Explain any two of them. [6]
 c) What do you mean by topic modelling? Explain Latent Dirichlet Allocation. [5]

❧ ❧ ❧